

NEPHROLOGY TEACHING SERIES

Hyponatremia

Tonicity, volume status, and the art of slow correction

Chronic hyponatremia is a water problem. Fix it slowly, or you'll break the brain.

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MS3/MS4 rotation teaching

Why we get consulted

Diagnosis looks simple; overcorrection isn't.

TRIGGERS FOR CONSULT

- Na < 125 or drop ≥ 10 in 24 h
- Symptomatic (seizure, AMS)
- Unclear etiology
- SIADH refractory to fluid restriction
- Cirrhosis / HF with progressive Na drop
- Overcorrection / DDAVP clamp for ODS

TEACHING PEARL

Three questions answer 90%: is it truly hypotonic? How concentrated is the urine? What's the volume status? Everything flows from these.

Three labs decide the diagnosis

Order all three on the same draw. Don't skip serum osmolality.

SERUM OSMOLALITY

Normal: 275–295 mOsm/kg

< 275 → TRUE hypotonic hyponatremia — proceed

275–295 → isotonic (pseudohyponatremia from lipids/protein)

> 295 → hypertonic (hyperglycemia, mannitol)

URINE OSMOLALITY

< 100 → water excess: primary polydipsia, beer potomania, tea & toast

100–300 → mixed / resetting

> 300 → ADH active: SIADH, hypovolemia, HF, cirrhosis, hypothyroid, adrenal insufficiency

URINE SODIUM

< 20 → low effective circulating volume (true hypovolemia or HF/cirrhosis)

> 30 → SIADH or renal salt loss (diuretics, CSW, adrenal insufficiency)

Helpful only after urine osm shows ADH is on

Etiology by volume status

Once true hypotonic, volume status sorts the differential.

HYPOVOLEMIC (true Na^+ loss > water loss)

- GI losses, diuretics (thiazides top offender), bleeding • Cerebral salt wasting (neuro patients) • Adrenal insufficiency / mineralocorticoid deficiency
- Urine $\text{Na} < 20$ (extrarenal); > 30 (renal) • Treat with isotonic saline → stop the ADH drive

EUVOLEMIC (water excess, Na^+ normal)

- SIADH (CNS, pulmonary, meds, malignancy) • Hypothyroidism • Glucocorticoid deficiency • Low solute intake (tea & toast, beer potomania) • Primary polydipsia • Management differs: fluid restriction ± salt tabs ± vaptan

HYPERVOLEMIC (impaired free water excretion)

- Heart failure, cirrhosis, nephrotic syndrome, advanced CKD • All share low effective arterial blood volume → ADH on despite expansion • Fluid + Na^+ restriction; treat underlying disease • Tolvaptan in hospitalized HF / cirrhosis with caution

SIADH — the diagnosis of pattern recognition

A clinical diagnosis that requires every exclusion before you commit.

DIAGNOSTIC CRITERIA (Bartter & Schwartz)

Hypotonic (serum osm < 275)

Urine osm > 100

Clinically euvolemic

Urine Na > 30

Normal TSH + cortisol

No recent diuretic

Responds to fluid restriction

CAUSES — 4 BUCKETS

CNS: stroke, SAH, trauma, meningitis

Pulmonary: pneumonia, TB, small-cell lung

Meds: SSRIs, carbamazepine, MDMA, thiazides

Malignancy (ectopic ADH)

Transient: post-op, pain, nausea

Correction rate — the central safety question

Goal: rise the Na by the right amount in the right time frame — no more.

ACUTE (< 48 h or severe symptoms)

Seizure, coma, respiratory arrest = brain swelling

3% saline 100 mL bolus IV over 10 min (may repeat ×2 until symptoms resolve)

Goal: raise Na^+ by 4–6 mEq/L quickly

Max 10–12 mEq/L in 24 h, 18 mEq/L in 48 h

Use 3% infusion 0.5–2 mL/kg/h after bolus

CHRONIC (> 48 h or unknown duration)

Brain has adapted — correcting fast causes ODS

Target: ≤ 8 mEq/L per 24 h (6 if high risk: alcoholism, malnutrition, hypoK)

Check Na^+ every 2 h initially

If overshooting: DDAVP 2–4 μg IV q6h + D5W

'DDAVP clamp' — pre-empt overcorrection in high-risk patients

Treatment — match the tool to the phenotype

Volume status drives the choice as much as the Na^+ value.

Phenotype	First-line	Second-line	Avoid
Severe symptomatic (acute)	3% saline 100 mL bolus × up to 3	Continuous 3% at 0.5–2 mL/kg/h; re-check q2h	<i>Isotonic saline (may worsen if SIADH)</i>
SIADH (euvolemic)	Fluid restriction 800–1000 mL/day	Salt tabs 1–2 g TID ± loop diuretic; selected inpatient tolvaptan if refractory	<i>Isotonic saline (desalination can worsen Na^+)</i>
Hypovolemic	Isotonic saline cautiously	Monitor for brisk overcorrection once volume restored	<i>Fluid restriction (worsens ADH drive)</i>
Hypervolemic (HF/cirrhosis)	Fluid + Na^+ restriction; loop diuretic	HF: selected inpatient tolvaptan. Cirrhosis/liver disease: generally avoid or use only with hepatology input	<i>Large volumes of saline; vaptans in liver failure</i>
Drug-induced (thiazide, SSRI)	Stop the drug	Fluid restriction until Na normalizes; expect brisk rise → DDAVP if overcorrecting	<i>Resuming thiazide without reassessment</i>

Osmotic demyelination syndrome (ODS)

The preventable complication that defines the field.

HIGH-RISK PATIENTS

- Baseline Na^+ < 120
- Chronic hyponatremia (unknown duration = assume chronic)
- Alcohol use disorder
- Malnutrition / cachexia
- Hypokalemia (K^+ repletion will drive Na^+ up further)
- Liver disease / advanced cirrhosis
- Hypoxia

PREVENTION / RESCUE

Prevention: limit Na^+ rise to $\leq 8 \text{ mEq/L/24 h}$ (6 in high risk)

Check Na^+ q2h during active correction

Replete K^+ into the Na budget (K rises raise Na too)

Rescue: if overshooting, DDAVP 2–4 μg IV + D5W 3 mL/kg over 1 h

Goal: re-lower Na^+ to stay under threshold

Clinical ODS may not appear for 2–6 days — act on labs, not symptoms

Common pitfalls

These are the ones that lead to ODS or to missed diagnoses.

AVOID

- Skipping serum osmolality
- Calling SIADH without TSH / cortisol
- Isotonic saline in true SIADH
- Fluid restriction in hypovolemia
- Not checking K (K raises Na)
- Correcting without Na q2h

DO INSTEAD

- Always order: serum osm, urine osm, urine Na
- Rule out thyroid / adrenal first
- SIADH → fluid restriction, not saline
- Hypovolemia → saline, watch rebound
- Budget K repletion into 24-h Δ Na
- q2h labs until stable

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 74-year-old woman with small-cell lung cancer presents with 3 days of progressive confusion. Na⁺ 114 (baseline 138 one month ago), no seizures. Exam: alert but disoriented × 2. Euvolemic. Serum osm 244, urine osm 560, urine Na 48. TSH and AM cortisol normal. On home sertraline; no diuretic. On admission she's given 500 mL NS.

QUESTION: What's the diagnosis, is this acute or chronic, and what's the correction target?

Clinical case — answer

Key teaching points from the case.

A N S W E R

SIADH secondary to small-cell lung cancer; chronic hyponatremia (> 48 h); target Na⁺ rise ≤ 6–8 mEq/L per 24 h to avoid ODS.

T E A C H I N G

Classic SIADH: hypotonic hyponatremia, inappropriately concentrated urine, urine Na > 30, euvolemic, normal TSH/cortisol. Isotonic saline can worsen SIADH via desalination. Treatment: fluid restriction (800-1000 mL/day) ± salt tabs; consider inpatient tolvaptan only if refractory and closely monitored. Na⁺ q2h during correction. If overcorrecting, rescue with DDAVP 2-4 µg IV + D5W 3 mL/kg. High-risk for ODS: alcoholism, malnutrition, hypokalemia, Na⁺ < 120.

REFERENCES

Hyponatremia

Guidelines

Expert panel consensus (Am J Med 2013) on hyponatremia
European clinical practice guideline — ERA-EDTA / ESE 2014
Endocrine Society — SIADH guidance
Hoorn & Zietse. JASN 2017 — practical correction approach

Landmark Trials

SALT-1 / SALT-2 — tolvaptan in SIADH, HF, cirrhosis
SALTWATER — long-term tolvaptan safety
Sterns et al. — ODS risk and correction pacing
Garrahy et al. — empiric fluid restriction success rates

UpToDate (2026 access):

Causes of hyponatremia in adults • Overview of the treatment of hyponatremia in adults • Treatment of hyponatremia: SIADH and reset osmostat • Osmotic demyelination syndrome and overly rapid correction